9/19/2025 Briana-Yves 1-1 Meeting Minutes

Craniopharyngiomas that involve the third ventricle, optic apparatus, or hypothalamus often cause severe endocrine dysfunction; in some cases, recurrence reduction strategies require pituitary gland removal with increased morbidity
Using manual segmentation, I presented several segmentation-based imaging metrics that could be examined to quantify tumor location and spatial relationships to these structures, and might help reveal associations between imaging phenotype and endocrine complications or resection outcomes.
Preliminary analysis indicated that the superior–inferior (SI) vertical offset (relative to the ventricular midline/AC–PC plane) was the most informative location metric and was prioritized for further study.
Computed radiomics features, including first-order, shape, GLCM, GLRLM, GLSZM, NGTDM, GLDM, and wavelet-derived features, might be used for correlation analysis with clinical variables such as hypothyroidism or Central Diabetes Insipidus.

Table 1 & 2: Imaging metrics

Supplement: metric definitions

case_id	surface_area_cm2	volume_cm3	elongation	sphericity	compactness	height_cm	depth_cm
71899681	49.77	24.52	1.098	0.820	0.005	3.6	3.9
76686586	120.59	73.78	1.322	0.705	0.003	5.1	7.3
77364735	37.10	15.34	1.225	0.805	0.005	3.5	3.9
77402268	129.96	71.28	1.323	0.640	0.002	6.9	6.3
79391843	42.60	16.25	1.416	0.728	0.003	4.2	3.9

case_id	aspect_ratio	ap_offset_mm	ap_position	vertical_offset_mm	midline_offset_mm	region_heuristic	
71899681	1.08	-0.33	near_midline	0.19	0.42	Midline	
76686586	1.43	0.15	near_midline	-0.11	0.31	Midline	
77364735	1.11	0.29	near_midline	-0.16	0.10	Midline	
77402268	0.91	0.10	near_midline	0.31	0.03	Midline	
79391843	0.93	0.24	near_midline	-0.27	0.43	Midline	

Table 1 & 2: Imaging Metric computed with a few available case IDs

Metric definitions

The elongation is computed as:

$$\text{Elongation} = \sqrt{\frac{\lambda_{\text{max}}}{\lambda_{\text{min}}}}$$

- If the tumor is roughly spherical, $\lambda_{\text{max}} \approx \lambda_{\text{min}}$, so elongation ≈ 1.
- If the tumor is stretched along one axis, $\lambda_{\text{max}} \gg \lambda_{\text{min}}$, so elongation becomes large.

$$\text{Sphericity} = \frac{\pi^{1/3}\,(6V)^{2/3}}{A}$$

- Sphericity ≈ 1 for a perfect sphere.
- Values decrease below 1 as the tumor becomes less spherical (e.g., irregular or elongated shapes).

$$\text{Compactness} = \frac{V^2}{A^3}$$

- Compactness is highest for sphere-like tumors.
- Values decrease toward 0 for shapes with large surface area relative to their volume (thin, elongated, or highly irregular tumor).
- V: tumor volume
- A: surface area of the tumor

$$\text{Aspect Ratio} = \frac{\text{AP}}{\text{SI}}$$

- If ≈ 1, the tumor is about as tall (SI) as it is deep (AP).
- If Aspect Ratio > 1.1, the tumor is elongated front–back compared to top–bottom.

$$d_{\text{midline}} = \left| c_x - c_{0x} \right|$$

- c_x: x–coordinate of the centroid (world coordinates).
- c_{0x}: x–coordinate of the image center (world coordinates).

$$d_{\mathrm{AP}} = \|\mathbf{c} - \mathbf{c}_0\| \cos(\theta_{\mathrm{AP}}) \cdot s_{\mathrm{AP}}$$

- d_{AP}: signed displacement along the anteroposterior (AP) axis.
- $\mathbf{c}$: tumor centroid in world coordinates.
- $\mathbf{c}_0$: image center in world coordinates.
- θ_{AP}: angle between displacement vector $(\mathbf{c} - \mathbf{c}_0)$ and the AP axis unit vector $\mathbf{u}_{\mathrm{AP}}$.
- s_{AP}: sign factor ($+1$ posterior, -1 anterior).

Similarly, d_{SI}: signed displacement along the superior-inferior (SI) axis.